I. Introduction

Foundations of Big Data

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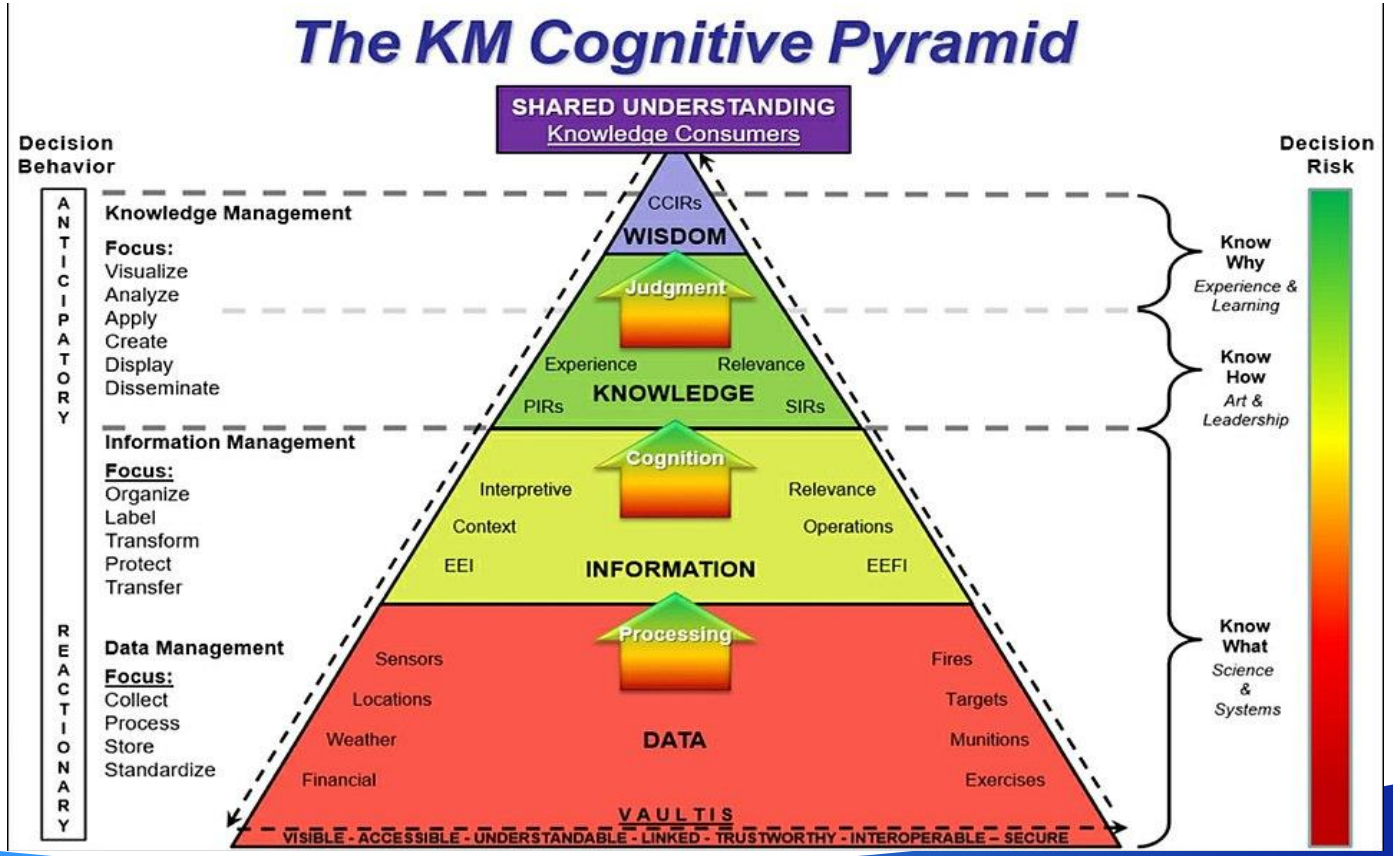
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Outline

1. Concepts: From Data to Intelligence
2. When Does Data Become “Big”?
3. Big Data Characteristics: 3Vs → System Requirements
4. The Big Data System: Pipeline, Architecture, and Lifecycle
5. Big Data Jobs

1. From Data to Intelligence (DIKW)

- Data → Information → Knowledge → Wisdom
- Reframed as pipeline: Data Collection → Processing → Analytics → Decision



What is Data?

- Raw, unprocessed facts or signals
- Without context or interpretation
- Examples:
  - 35°C
  - 120 bpm
  - 1,000,000 transactions
  - "click", "view", "purchase"
- Characteristics:
  - Structured / unstructured
  - No inherent meaning
  - High volume

Raw data collected from sensors:

- Timestamp: 08:00:02
- Vehicle count: 120
- Avg speed: 18 km/h
- Location: Intersection A

At this stage:

✓ No explanation.
✓ No conclusion.
✓ Just signals.

- In Big Data: Logs, sensor streams, images, text, transactions, …

What is Information?

- Data that has been:
  - Processed
  - Structured
  - Contextualized
  - Aggregated
- Information answers:

  → Who?
  → What?
  → When?
  → Where?
- Example:
  - "Traffic congestion detected at Intersection A during morning rush hour."
  - Now data has context.

From raw logs → aggregated metrics:

- Average speed dropped 40%
- Vehicle density exceeded threshold
- Peak congestion between 7:45–8:30 AM

Now we understand:

✓ There is congestion.
✓ But we do not yet know why.

What is Knowledge?

- Information combined with:
  - Experience
  - Patterns
  - Models
  - Analytical reasoning
- Knowledge answers:
  - → How?
  - → Why?

After analysis:

- Pattern repeats weekly
- Correlated with school start time
- Signal timing is suboptimal

We now understand:

✓ Cause–effect relationships.

This requires:

✓ Statistical analysis
✓ ML models
✓ Domain expertise

- Example:
  - "Congestion occurs every weekday due to nearby school traffic and insufficient signal timing."
  - Now we understand the mechanism.

What is Wisdom?

- Knowledge applied with:
  - Judgment
  - Long-term perspective
  - Strategic thinking
  - Ethical considerations
- Wisdom answers:

  → What should we do?
- Example:
  - "Adjust traffic signals dynamically and redesign road layout to reduce long-term congestion."

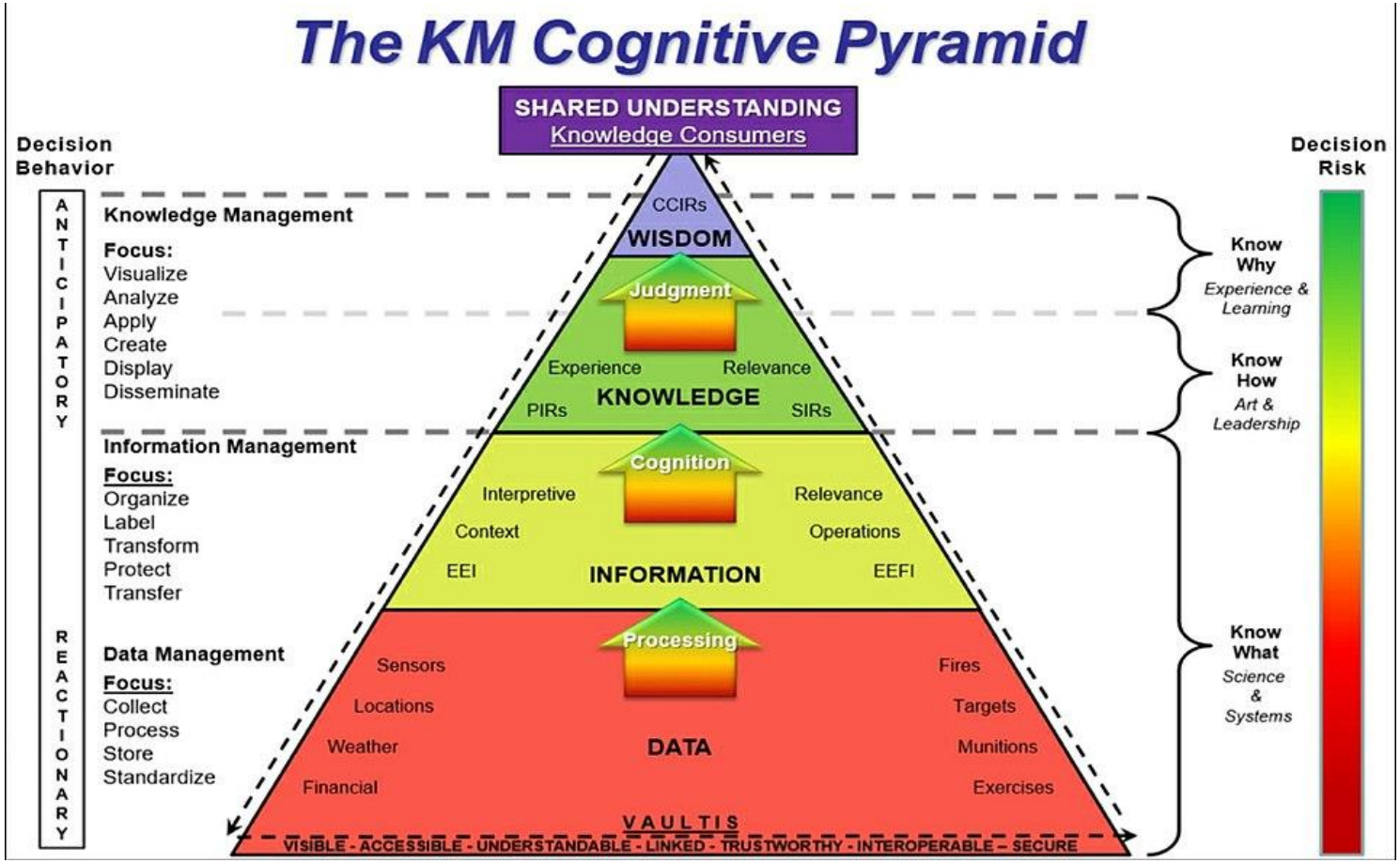
Decision:

- Deploy adaptive traffic control
- Introduce staggered school schedule
- Invest in infrastructure upgrade

Wisdom includes:

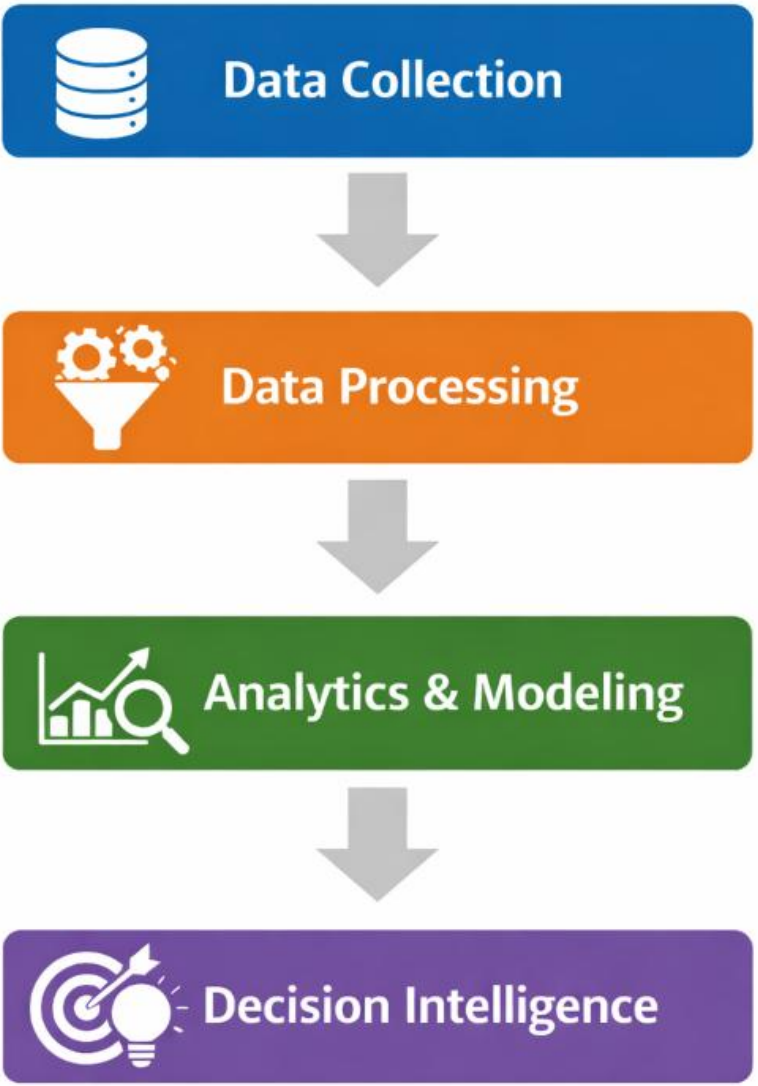
✓ Cost-benefit analysis
✓ Social impact
✓ Sustainability

Turning Data to Strategic Wisdom



- **Big Data engineering enables movement from D → W.**
- **AI accelerates K → W.**

DIKW Mapped to Systems



Level	System Component
Data	Distributed Storage
Information	ETL / Aggregation
Knowledge	ML / Analytics Engines
Wisdom	Decision Support Systems

Course Positioning

- **Big Data is NOT:**
  - ❌ Data Science
  - ❌ Artificial Intelligence
  - ❌ Just “Large Dataset”
- **Big Data IS: Engineering Scalable Distributed Data Systems**

Domain	Focus
Data Science	Insight, modeling
AI	Learning & reasoning
Big Data	Infrastructure & scalability

2. Why study big data?

- **Studying Big Data technologies means learning how to transform massive raw data into scalable intelligence and strategic decisions.**

Data is growing exponentially

- Massive volume, high velocity, high variety
- Traditional systems cannot scale

AI depends on it

- Large datasets, Distributed storage, Parallel computing
- 👉 No Big Data infrastructure → No scalable AI

Orgs compete on Intelligence

- Big Data enables: Real-time analytics, Predictive modeling, Data-driven decision making
- From **Data** → **Insight** → **Action** → **Competitive Advantage**

It is a core digital skill

- Data Engineer, Data Scientist, ML Engineer, Analytics Architect

When Systems Break

Common Failure Points

- 🧠 Out of Memory
- 💿 Disk I/O Bottleneck
- 🔥 CPU Saturation
- 🌐 Network Congestion
- 💥 Single Node Failure
- ⏳ Latency Explosion

Motivate to:

- Hadoop
- Distributed file systems
- Parallel computing

The Real Limitation

Single Machine Constraints

- Limited RAM
- Limited Disk Throughput
- Limited CPU cores
- Single Point of Failure

Big Data begins when single-node systems are no longer sufficient.

Scale-Up vs Scale-Out

- **Vertical Scaling (Scale-Up)**
  - Bigger server
  - Expensive
  - Hardware ceiling
  - Single failure risk

- **Horizontal Scaling (Scale-Out)**
  - Add more machines
  - Distributed workload
  - Fault tolerance
  - Elastic scalability

Distributed System Thinking

- A Distributed System =
  - Many machines
  - Network communication
  - Partial failures
  - Coordination challenges

- Physical Constraints:
  - Memory hierarchy differences
  - Disk vs RAM speed gap
  - Network latency
  - CPU parallelism limits

- Key Concepts:
  - Node
  - Cluster
  - Replication
  - Partitioning
  - Consensus

Big Data systems must optimize across these constraints.

How much data?

Google™
Processes 40 EB a day (2023)
Search Index 100 EB (2023)
Perform 8.5B searches/day (2023)
Crawls 20B web pages a day (2023)

YAHOO!®
19 Hadoop clusters: 600 PB, 40k servers (9/2015)

JPMorganChase
550 PB on 50k+ servers running 15k apps (2024)

ebay®
Hadoop: 10K nodes, 150K cores, 150 PB (4/2014)

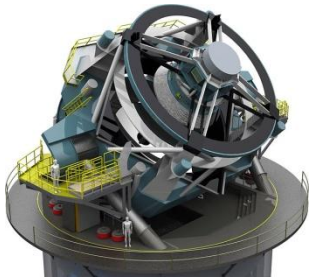
1,000 PB data in Hive + 4 PB/day (2024)

facebook

amazon web services™
S3: 2T objects, 1.1M request/second (4/2013)

LHC: ~15 PB a year

CERN



LSST: 6-10 PB a year (~2020)



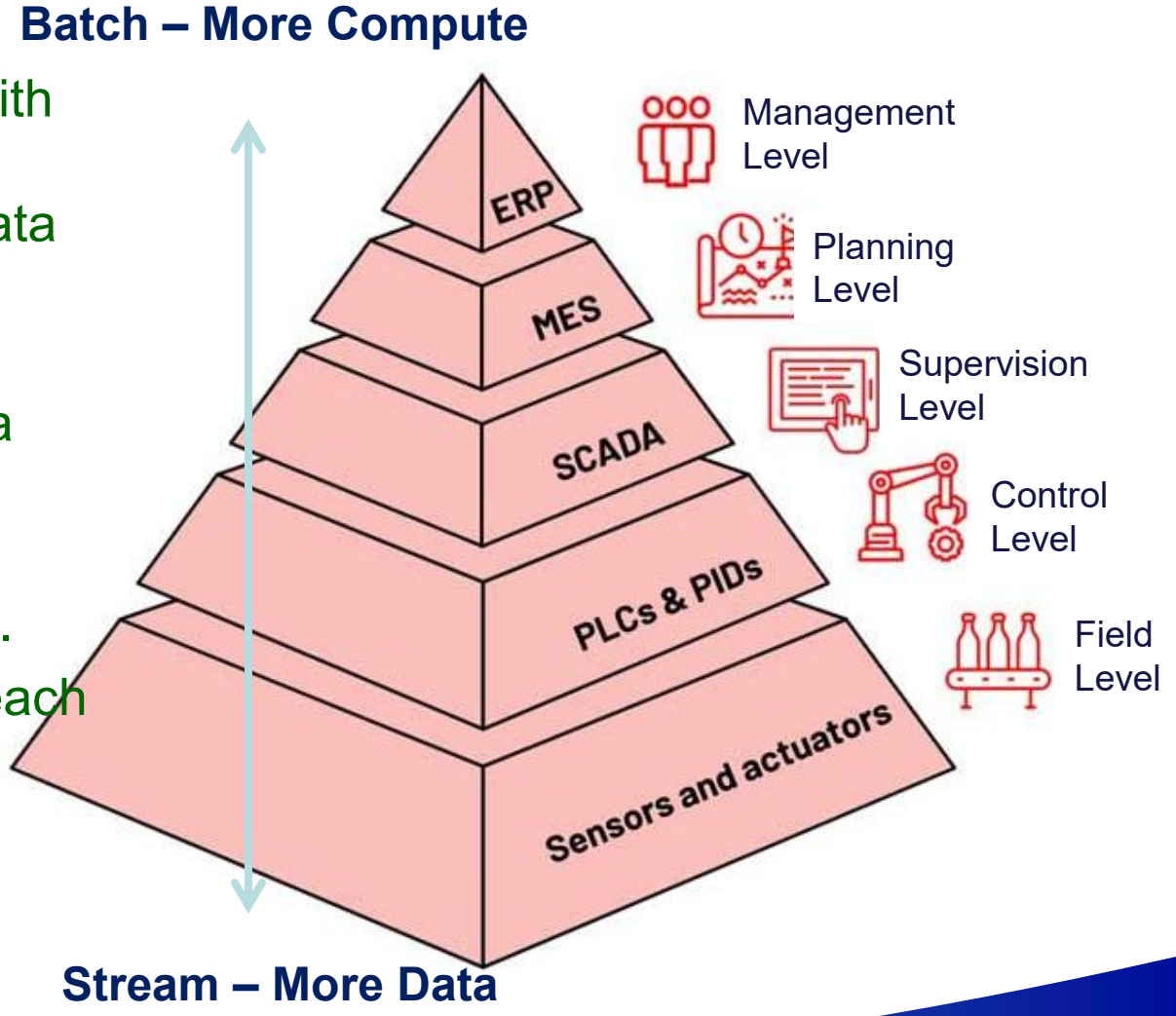
SKA: 0.3 – 1.5 EB per year (~2020)



From http://www.umiacs.umd.edu/~jimmylin/

How much data?

- Airbus A350: Equipped with ~6,000 sensors, generating ~2.5 TBs of data per day.
- Autonomous vehicles: Generate 4~6 TBs of data per day.
- Smart factories: Produce ~500 TBs of data per day.
- IoT data is projected to reach 73.1 ZB by 2025. (Source: IDC)



What is big data?

Big data is data that exceeds the processing capacity of conventional database systems. The data is too big, moves too fast, or doesn't

Big data is the data characterized by 3 attributes: volume, variety and velocity.

- IBM

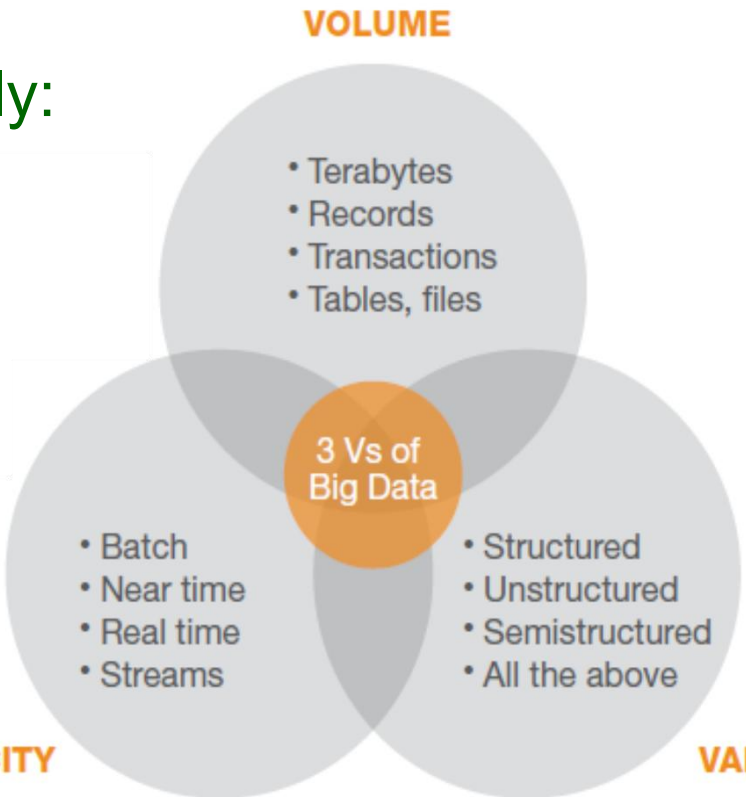
Big data is the data characterized by 4 key attributes: volume, variety, velocity and value.

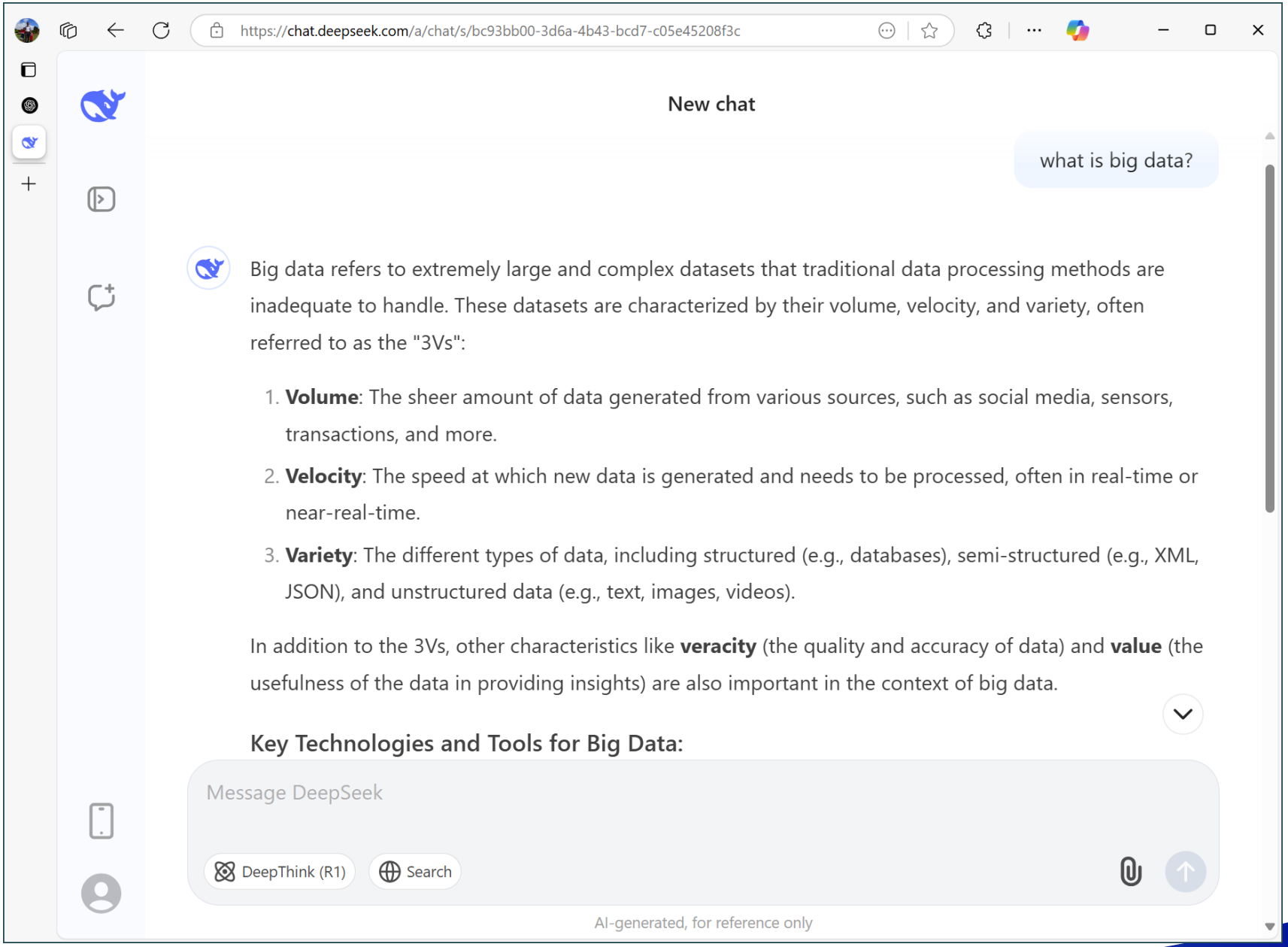
- Oracle

ORACLE®

Big Data

- **Big** in Big Data refers to:
  - Big **size** is the primary definition.
  - Big **complexity** rather than big volume. it can be small and not all large datasets are big data
  - size matters... but so does **accessibility, interoperability** and **reusability**
- define Big Data using 3Vs; namely:
  - **Volume, Variety, Velocity**

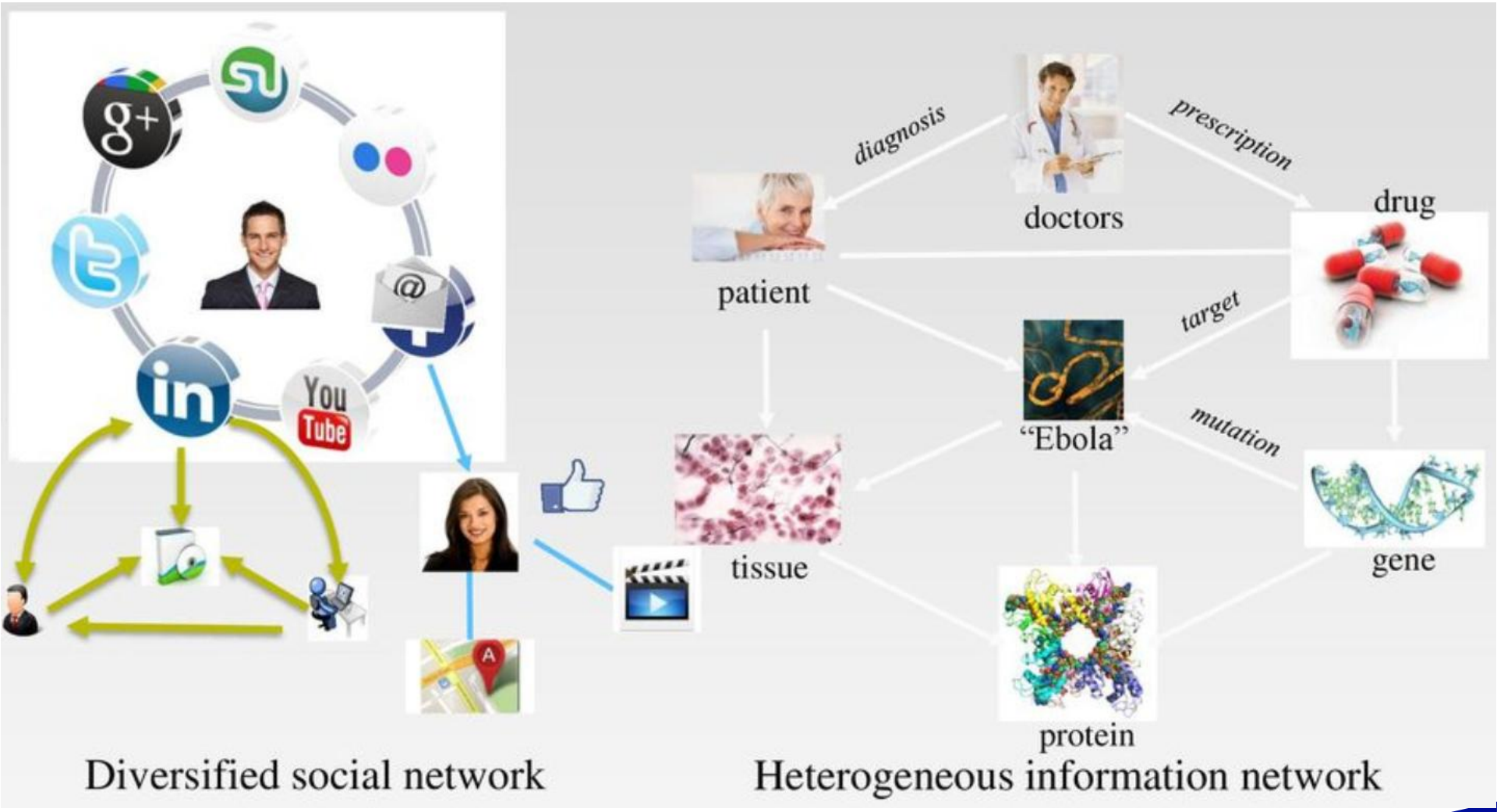



https://chat.deepseek.com/a/chat/s/bc93bb00-3d6a-4b43-bcd7-c05e45208f3c
New chat
what is big data?
Big data refers to extremely large and complex datasets that traditional data processing methods are inadequate to handle. These datasets are characterized by their volume, velocity, and variety, often referred to as the "3Vs":
1. **Volume**: The sheer amount of data generated from various sources, such as social media, sensors, transactions, and more.
2. **Velocity**: The speed at which new data is generated and needs to be processed, often in real-time or near-real-time.
3. **Variety**: The different types of data, including structured (e.g., databases), semi-structured (e.g., XML, JSON), and unstructured data (e.g., text, images, videos).
In addition to the 3Vs, other characteristics like **veracity** (the quality and accuracy of data) and **value** (the usefulness of the data in providing insights) are also important in the context of big data.
Key Technologies and Tools for Big Data:
Message DeepSeek
DeepThink (R1)
Search
AI-generated, for reference only

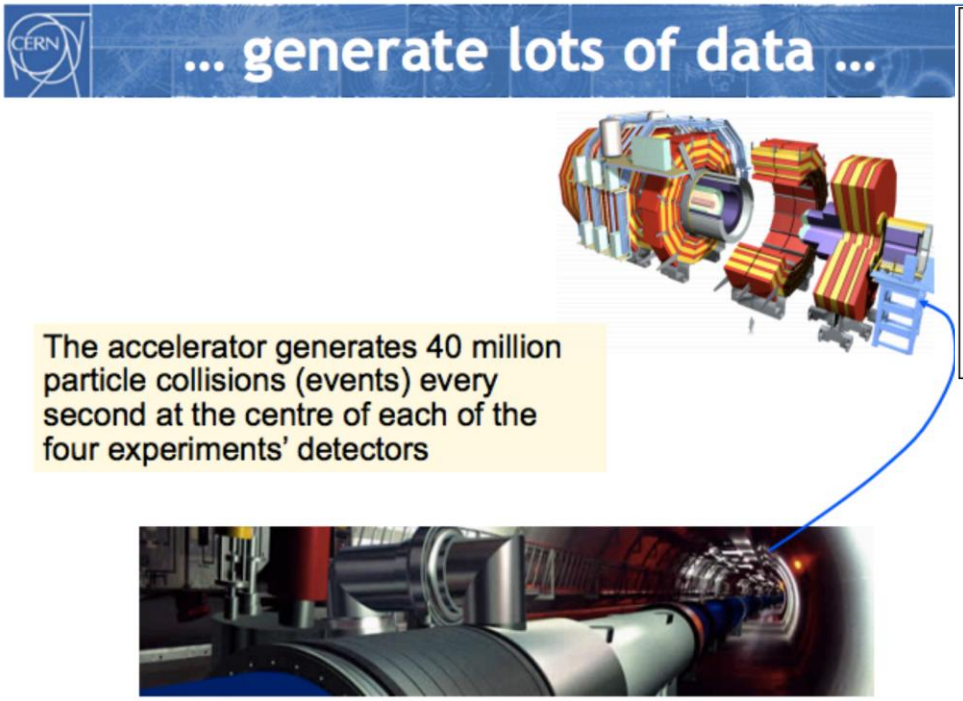
Why Big Data Matters

- Transforms raw information into insights and competitive advantage
- Enables smarter decisions, predictions, and automation
- Drives innovation in nearly every industry
- Helps organizations identify patterns impossible to see manually

A global view of linked big data



Realtime Analytics/Decision Requirements





- Today, it is possible using real-time analytics to optimize Like buttons across both website and on Facebook.
- FaceBook use anonymised data to show the number of times people:
  - saw Like buttons,
  - clicked Like buttons,
  - saw Like stories on Facebook,
  - and clicked Like stories to visit a given website.

Business Value of Big Data

- Improved customer experience (personalization, recommendations)
- Optimized operations & cost reduction
- Fraud detection & enhanced security
- Data-driven product development
- Predictive maintenance & reliability

Real-world examples

- Finance: real-time fraud detection
- Healthcare: disease prediction, precision medicine
- Retail: personalized offers, inventory forecasting
- Transportation: route optimization, autonomous vehicles
- Smart Cities: energy monitoring, traffic management
- …

Big Data: A buzz word?



2008



2011



2012



2022



2010



2011



2018

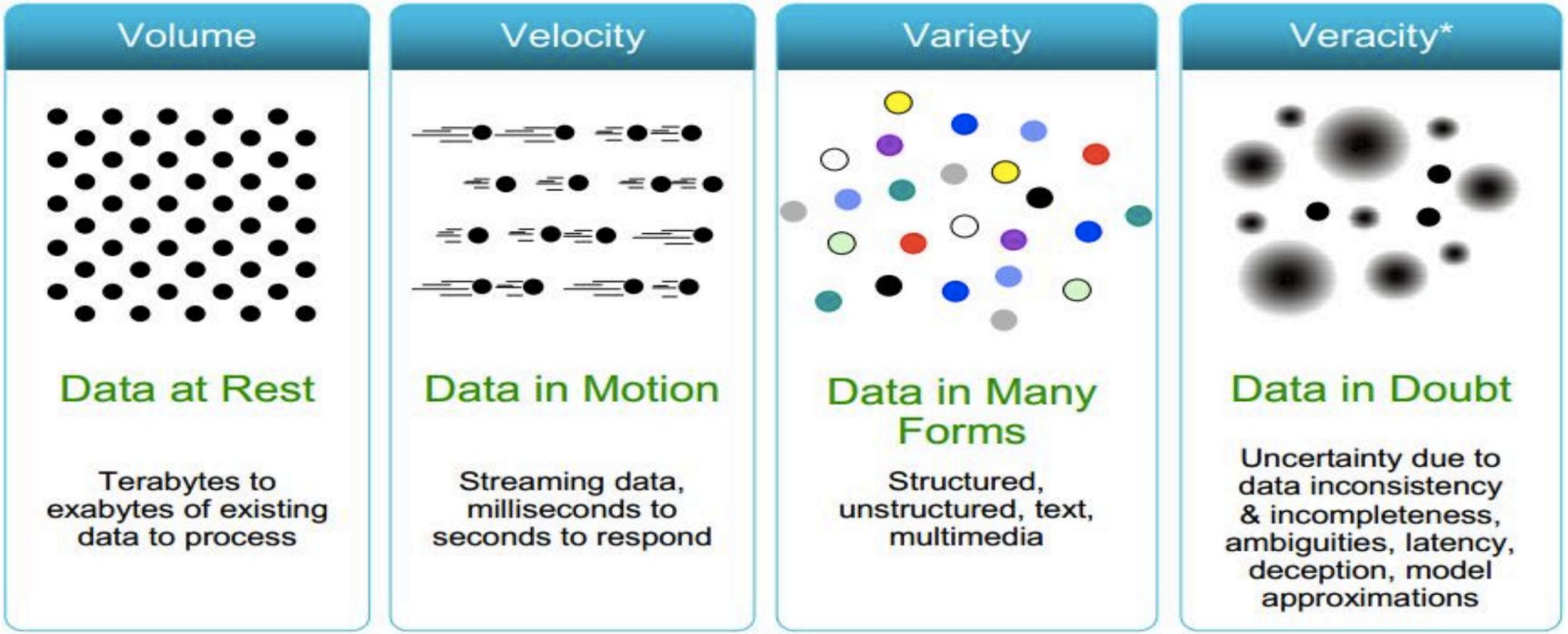
https://scholar.google.com.sg/citations?view_op=top_venues&hl=en&vq=eng_dataminninganalysis

Data Mining & Analysis

	Publication	h5-index	h5-median
1.	ACM SIGKDD Conference on Knowledge Discovery and Data Mining	124	213
2.	IEEE Transactions on Knowledge and Data Engineering	107	185
3.	International Conference on Artificial Intelligence and Statistics	100	162
4.	Journal of Big Data	79	191
5.	ACM International Conference on Web Search and Data Mining	77	130
6.	IEEE International Conference on Big Data	54	79
7.	ACM Conference on Recommender Systems	53	81
8.	Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery	51	113
9.	IEEE International Conference on Data Mining	50	88
10.	Knowledge and Information Systems	48	85
11.	ACM Transactions on Intelligent Systems and Technology (TIST)	47	85
12.	Data Mining and Knowledge Discovery	45	78
13.	European Conference on Machine Learning and Knowledge Discovery in Databases	45	69
14.	ACM Transactions on Knowledge Discovery from Data (TKDD)	44	67

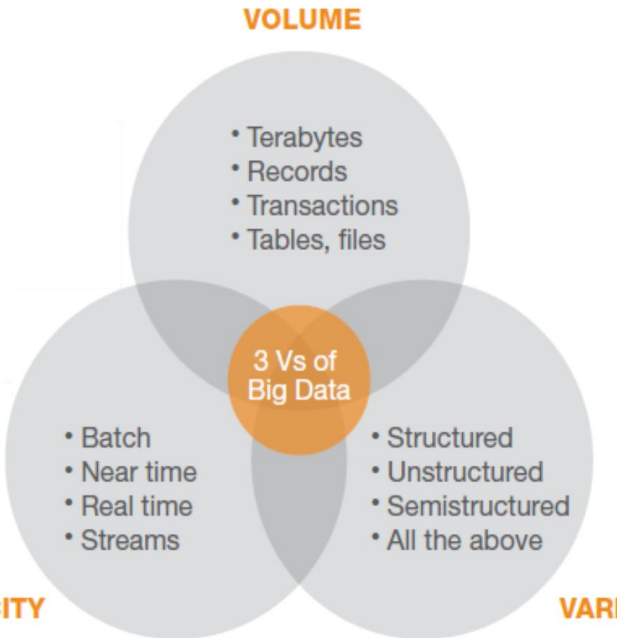
3. Big Data Characteristics

- Big Data is not specific application type, but rather a trend –or even a collection of trends- napping multiple application types
- Data growing in multiple ways
  - More data (volume of data)
  - More Type of data (variety of data)
  - Faster Ingest of data (velocity of data)
  - More Accessibility of data (internet, instruments , …)
  - Data Growth and availability exceeds organization ability to make intelligent decision based on it



Volume, Variety, and Velocity

- Aggregation that used to be measured in petabytes (PB) is now referenced by a term: zettabytes (ZB).
  - A zettabyte is a trillion gigabytes (GB)
  - or a billion terabytes
- in 2010, we crossed the 1 ZB marker, and at the end of 2025 that number was estimated to be 175 ZB (source IDC)
  - Google: 100 PB/day process, 15000 PB storage
  - eBay: 100 TB/day, 90 PB storage
  - Baidu: 10-100 TB/day, 2000 PB storage
  - Facebook: 600 TB/day, 300 PB storage
  - Spotify: 2.2 TB/day, 100 PB storage



Volume, Variety, and Velocity

- Different types: single application can be generating/collecting many types of data
  - Relational data (relation/transactions/legacy data)
  - Text data (Web)
  - Semi-structured data (XML)
  - Graph data: social network, semantic web, …
  - Streaming data, …
- Different sources:
  - User shopping behaviors from Shoppe/Lazada/Tiki/Amazon …
  - Product reviews from different provider websites

➔ To extract knowledge, all these types of data need to be linked together

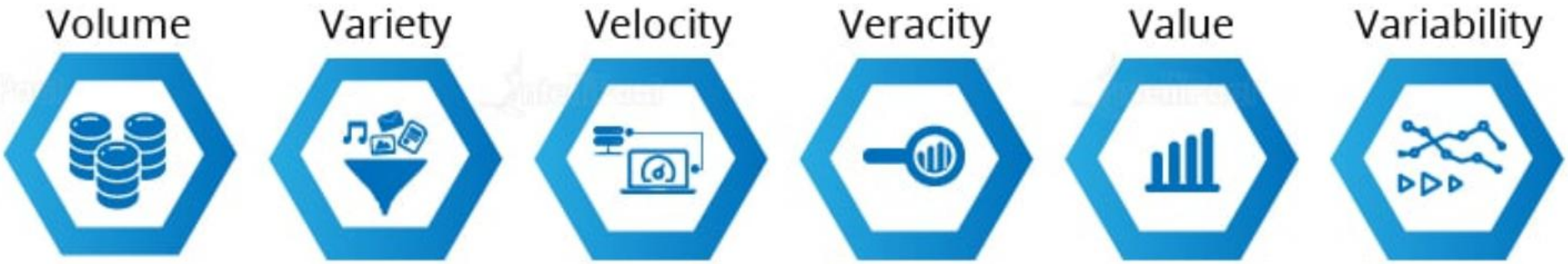
  - Trying to capture all of the data that pertains to our decision-making process.
  - Making sense out of unstructured data, such as opinion, or analysing images.

Volume, Variety, and Velocity

- The rate at which data arrives at the enterprise and is processed or well understood
- In other terms “How long does it take you to do something about it or know it has even arrived?
- Data is generated fast and need to be processed fast
- Late decisions ➔ missing opportunities

- Examples:
  - e-Promotions: based on your current location, your purchase history, what you like → send promotions right now for the store next to you
  - Healthcare monitoring: sensors monitoring your activities and body → any abnormal measurements require immediate reaction
  - Disaster management and response

Extensions: 6Vs of Big Data



- Volume: The amount of data being generated is growing rapidly and becoming increasingly large, making it difficult to store and process using traditional methods.
- Variety: Big data comes in many different formats, including structured, semi-structured, and unstructured data, such as text, images, videos, and sensor data.
- Velocity: Big data is generated and processed at a high speed, requiring real-time processing capabilities.
- Veracity: The quality of big data can be uncertain, making it important to validate and clean the data before using it for analysis.
- Value: Despite its challenges, big data holds the potential to deliver valuable insights and drive business results, making it an important asset for organizations.
- Variability: refers to how often this change happens. Big Data helps in managing these drifts of data that benefit organizations to come up with the latest products.

Other V's

- Visibility/Visualization: after big data being processed, we need a way to present the data in a manner that is readable and accessible.
- Viscosity: describe the latency or lag time in the data relative to the event being described. We found that this is just as easily understood as an element of Velocity.
- Virality: defined by some users as the rate at which the data spreads; how often it is picked up and repeated by other users or events.
- Volatility: refers to how long data is valid and how long it should be stored. You need to determine at what point data is no longer relevant to the current analysis.
- …

Who cares Big Data?

- Government
- Finance, Banking
- Manufacture
- Education
- Health
- Traffic
- IoT
- ….



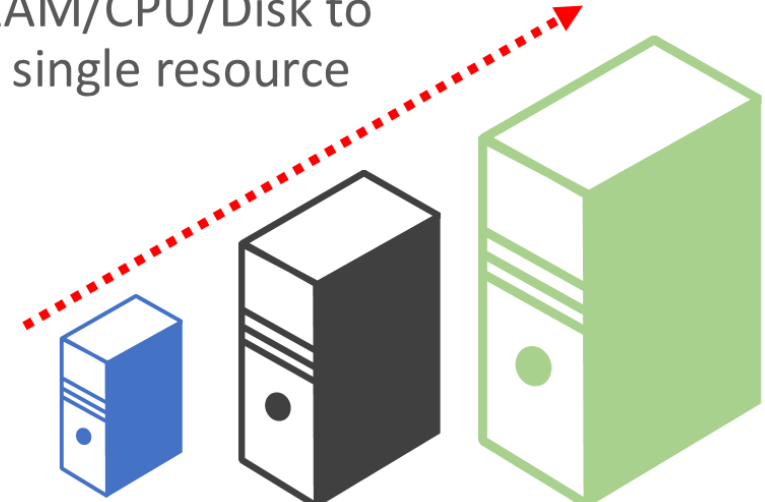
How to deal with Big Data?

Advice From Jim Gray (Turing 98):

1. Analysing big data requires **scale-out** solutions not **scale-up** solutions
2. Move the analysis to the data.
3. Work with scientists to find the most common **"20 queries"** and make them fast.
4. Go from "working to working."

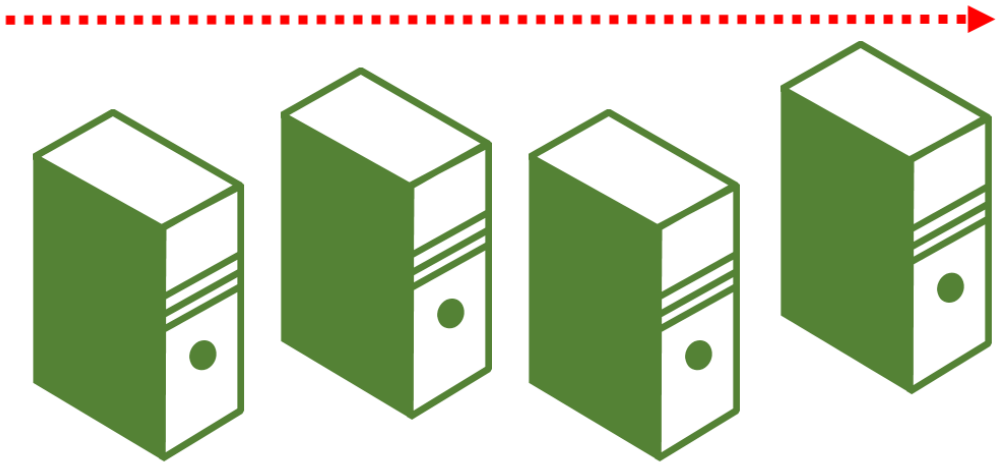
Scale Up (vertical scaling)

Increase capacity by adding RAM/CPU/Disk to a single resource



Scale Out (horizontal scaling)

Increase capacity by adding resources



20 Common Queries

1. Data retrieval & search

- **Find a specific record by ID** (e.g., retrieve a user, product, or experiment result).
- **Retrieve a list of recent records** (e.g., latest 100 transactions, newest publications).
- **Full-text search on large text fields** (e.g., search scientific papers for a keyword).
- **Find records with partial matches** (e.g., autocomplete suggestions for search terms).

2. Aggregation & Statistics

- **Compute total count** (e.g., how many users made purchases this month?).
- **Find the sum, average, min, max of a field** (e.g., average sensor temperature, highest sales).
- **Group by and aggregate** (e.g., total sales per region, number of cases per category).

3. Time-Based Analysis

- **Find records in a given time range** (e.g., transactions from Jan 1–Feb 1).
- **Compare trends over time** (e.g., sales growth from last year to this year).
- **Compute moving averages over time** (e.g., rolling 7-day average of website visits).

4. Join & Relationships

- **Join two or more datasets** (e.g., link customer data with purchase history).
- **Find co-occurring items** (e.g., products frequently bought together).
- **Detect missing or mismatched records** (e.g., students registered for a course but missing from attendance logs).

5. Ranking & Sorting

- **Find the top N results** (e.g., top 10 best-selling books).
- **Find anomalies/outliers** (e.g., detect unusually high spending in credit card transactions).

7. Geospatial Processing

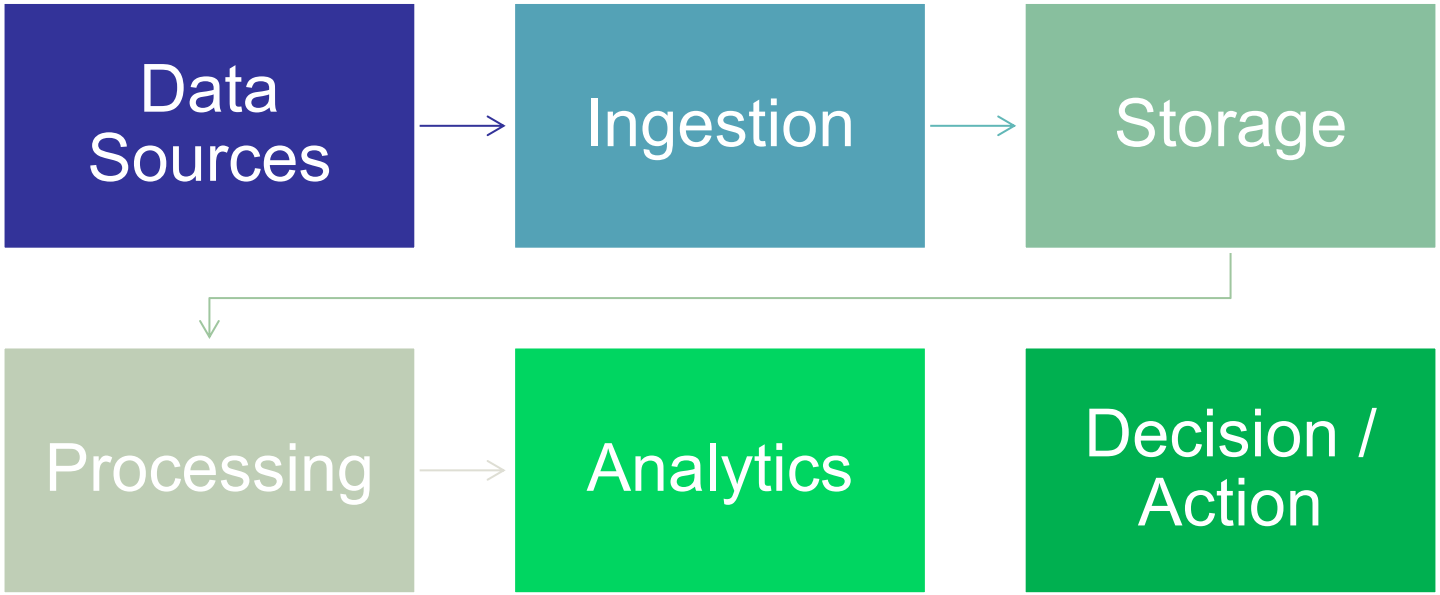
- **Find records within a location radius** (e.g., all hospitals within 10km of a given point).
- **Find the nearest neighbors** (e.g., closest weather stations to a given location).

8. Machine Learning & Anomaly Detection

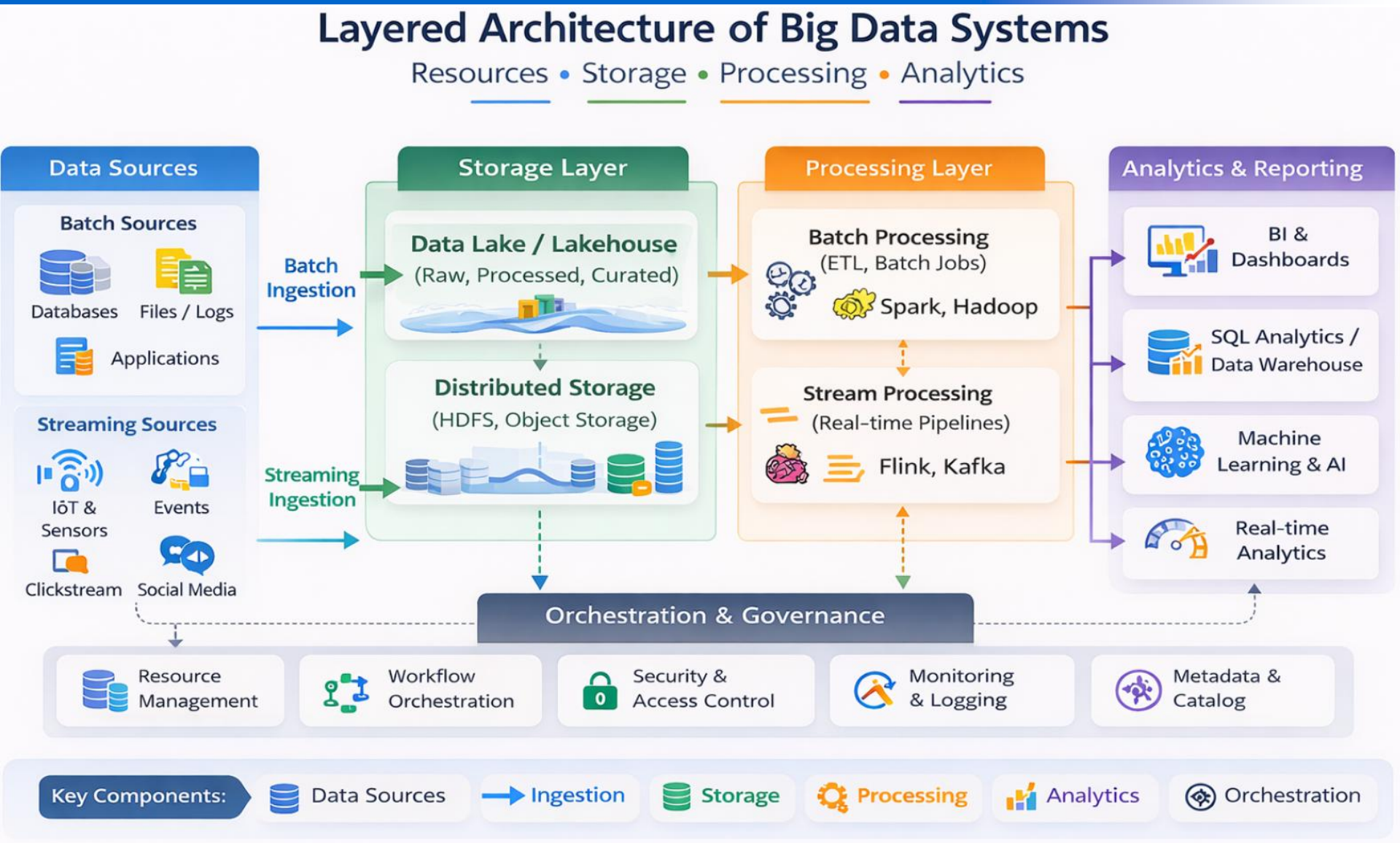
- **Find similar records** (e.g., customers with similar purchasing behavior).
- **Detect duplicate records** (e.g., merge duplicate user profiles).
- **Find sudden spikes or drops** (e.g., unusual traffic surge on a website).
- **Detect missing expected values** (e.g., expected daily report missing for a date).

4. The Big Data System

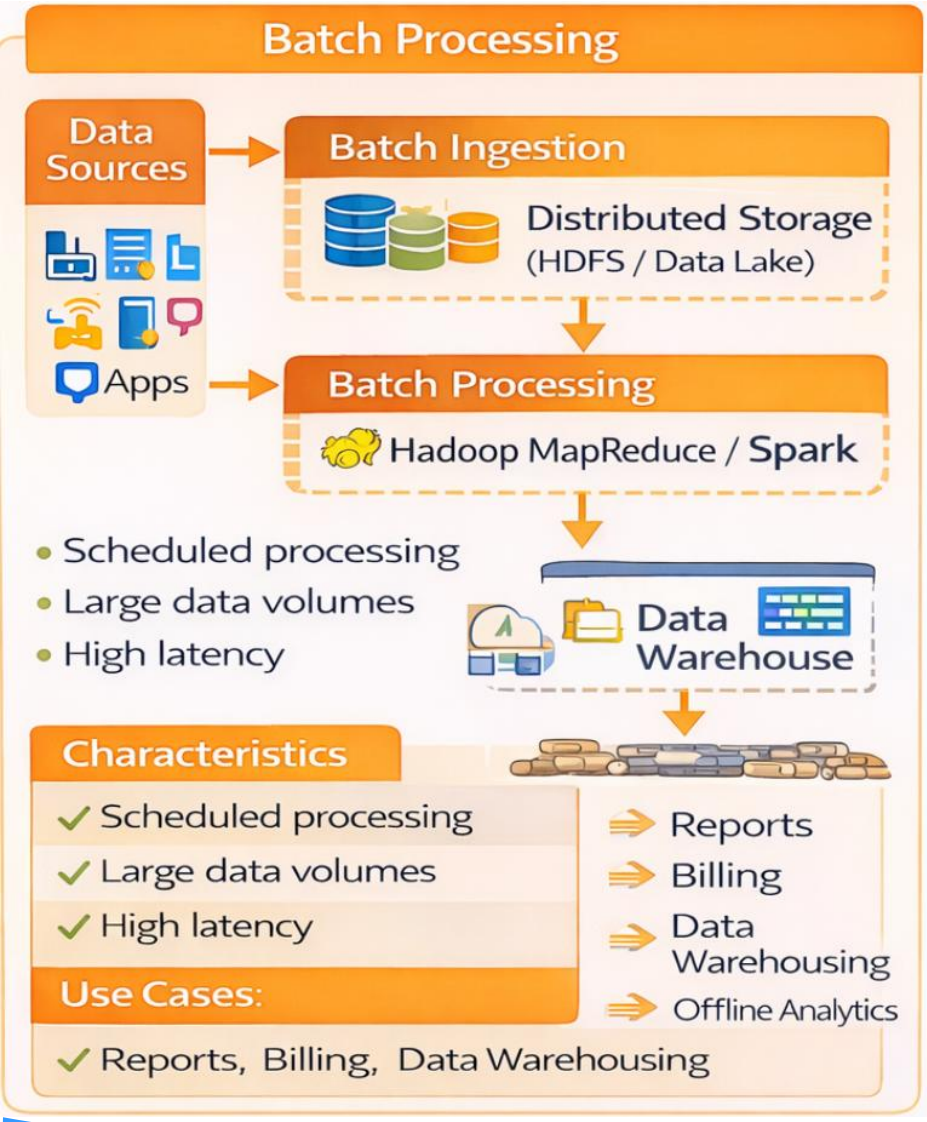
- Big Data System Model: Value Chain – Architecture - Life Cycle
- Big Data Value Chain
  - Focus: data → insight → business value
  - Viewpoint: business + system architecture

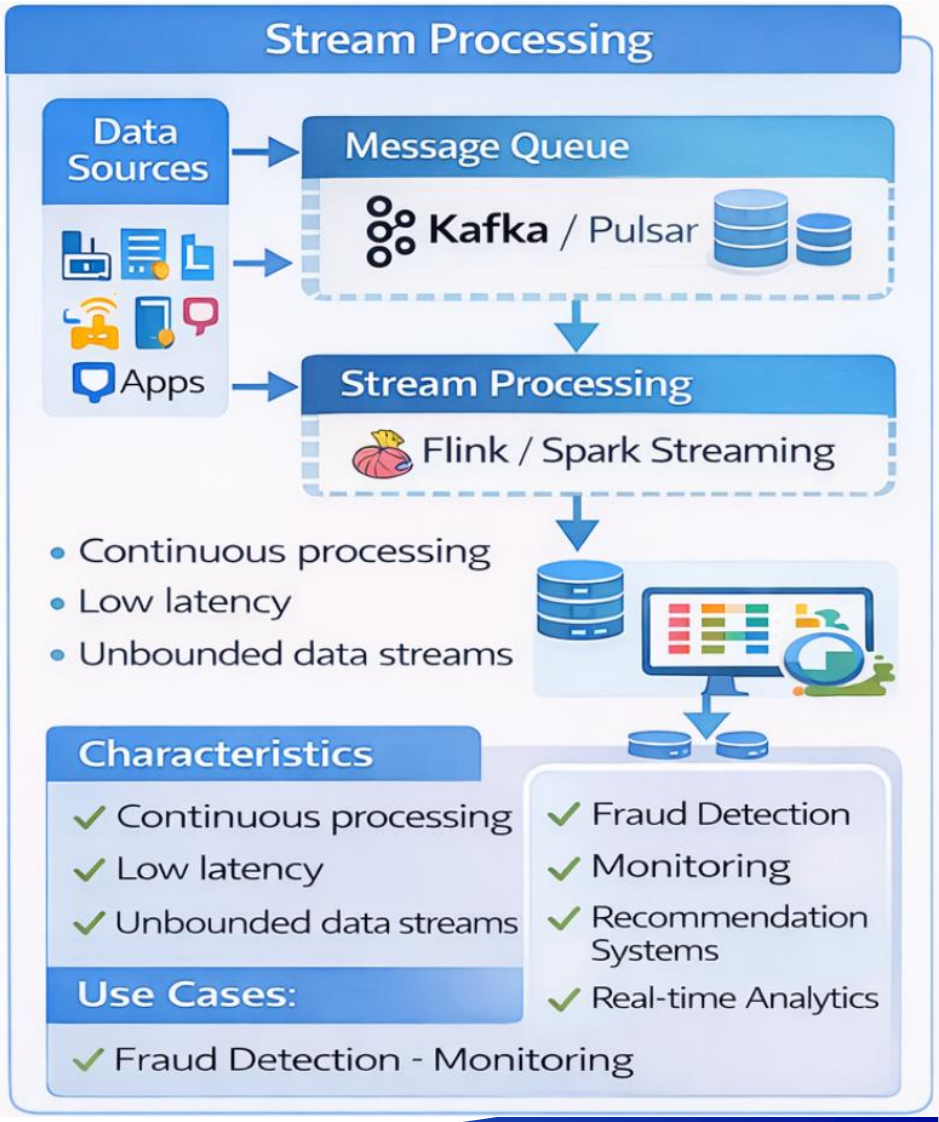


Big Data Architecture



Batch vs Streaming Processing





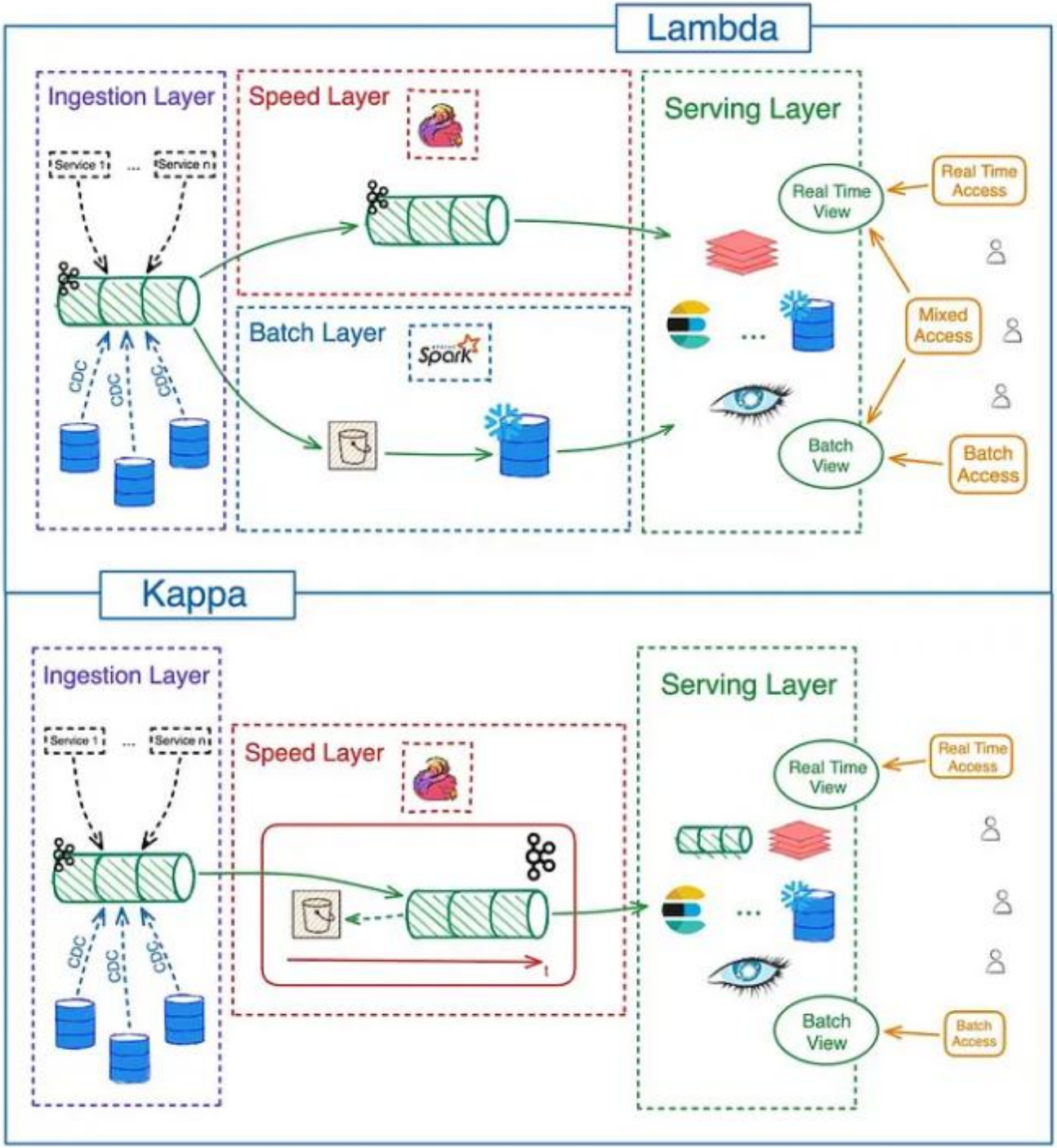
Lambda vs Kappa Architecture

- **Idea:** combination of **batch processing + stream processing**
- **Characteristics:** 2 pipeline:
  - **Batch layer** → historical data
  - **Speed layer** → real-time data

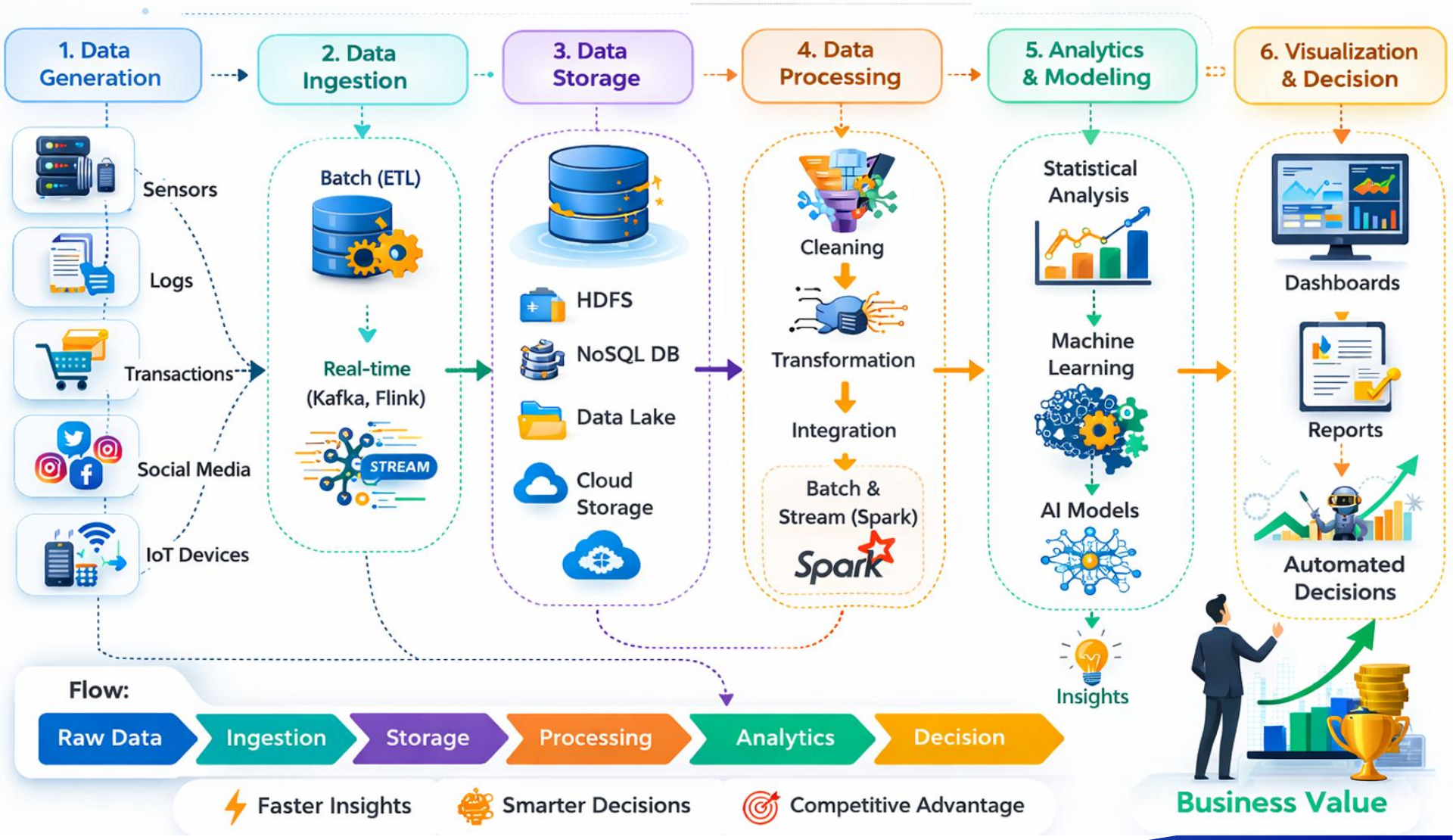
results produced in **serving layer**

- **Advantage:** accurate, scalable, fault tolerant
- **Disadvantage:** complex, maintain **2 codebases**

- **Idea**: only **stream processing**
- **Characteristics:** single pipeline:
  - **historical data = replay stream**
- **Advantage:** simple, single codebase
- **Disadvantage:** dependency of stream engine performance


Lambda
Ingestion Layer
Service 1
...
Service n
CDC
CDC
CDC
Speed Layer
Batch Layer
Spark
Serving Layer
Real Time View
Real Time Access
Mixed Access
Batch View
Batch Access
Kappa
Ingestion Layer
Service 1
...
Service n
CDC
CDC
CDC
Speed Layer
t
Serving Layer
Real Time View
Real Time Access
Batch View
Batch Access

Big Data Life Cycle



Characteristics



- Iterative, Not Linear
  - Feedback loops refine models
  - Continuous data ingestion
  - Continuous model retraining

- Key Engineering Challenges
  - Scalability (horizontal scale-out)
  - Fault tolerance
  - Data quality
  - Governance & security
  - Latency (real-time vs batch)

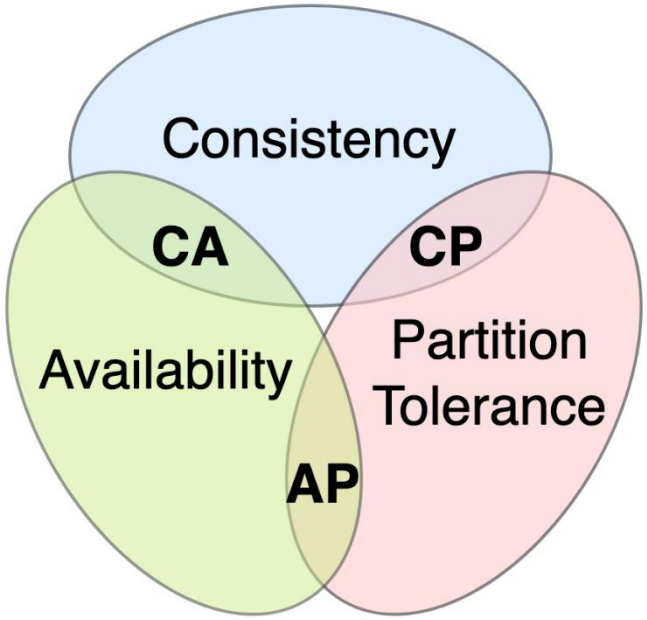
What Big Data Engineering Builds

- Distributed storage systems
- Parallel processing engines
- Resource management
- Data governance
- Fault-tolerant infrastructure

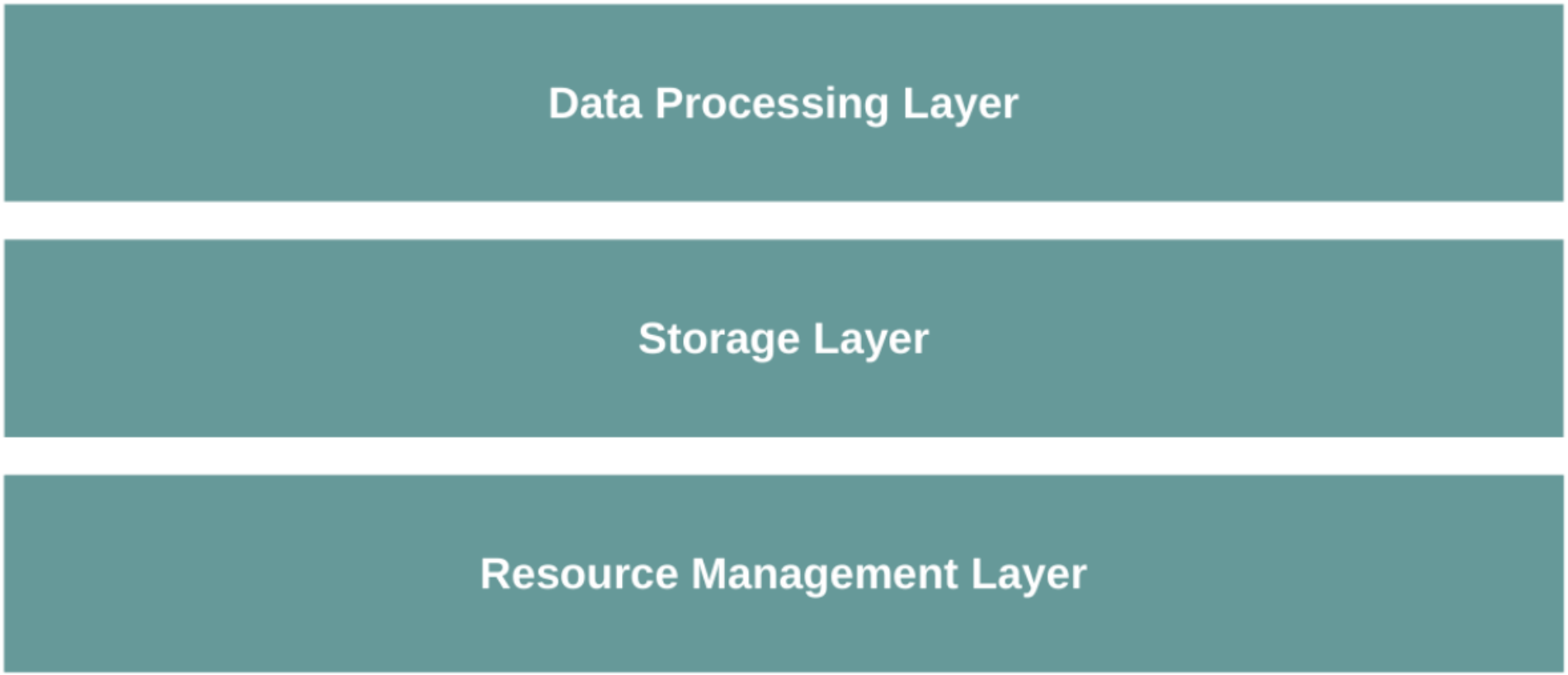
CAP Theorem (Conceptual)

- You can only guarantee two of:
  - Consistency
  - Availability
  - Partition Tolerance

- Design = Choosing trade-offs.



Target stack on this course



Resource Management Layer

- Cluster scheduling
- Resource allocation
- Failure handling
- Monitoring

Examples:

- YARN
- Kubernetes

Storage Layer

- HDFS
- Object Storage
- Replication
- Data Locality

Goals:

- Scalability
- Reliability
- Cost Efficiency

Processing Layer

- Batch Processing
- Stream Processing
- Iterative Processing

Examples:

- MapReduce
- Spark
- Flink

Example: Word Count at Scale

- Laptop:
  - Single CPU
  - Limited RAM
  - Sequential execution

- Cluster:
  - Many nodes
  - Parallel execution
  - Fault tolerance

When Do You Need Big Data?

- Data too large for one machine
- Real-time requirements
- High concurrency
- Complex distributed ML workloads

When NOT to Use Big Data

- Small datasets
- Low scalability need
- Simple workloads
- Over-engineering risk

6. Big Data Jobs

- Data scientists: collect, analyze, manage, structure and interpret large volumes of data from a range of sources. Data scientists then use reporting tools to pinpoint patterns, trends and interrelationships between the various data sets.

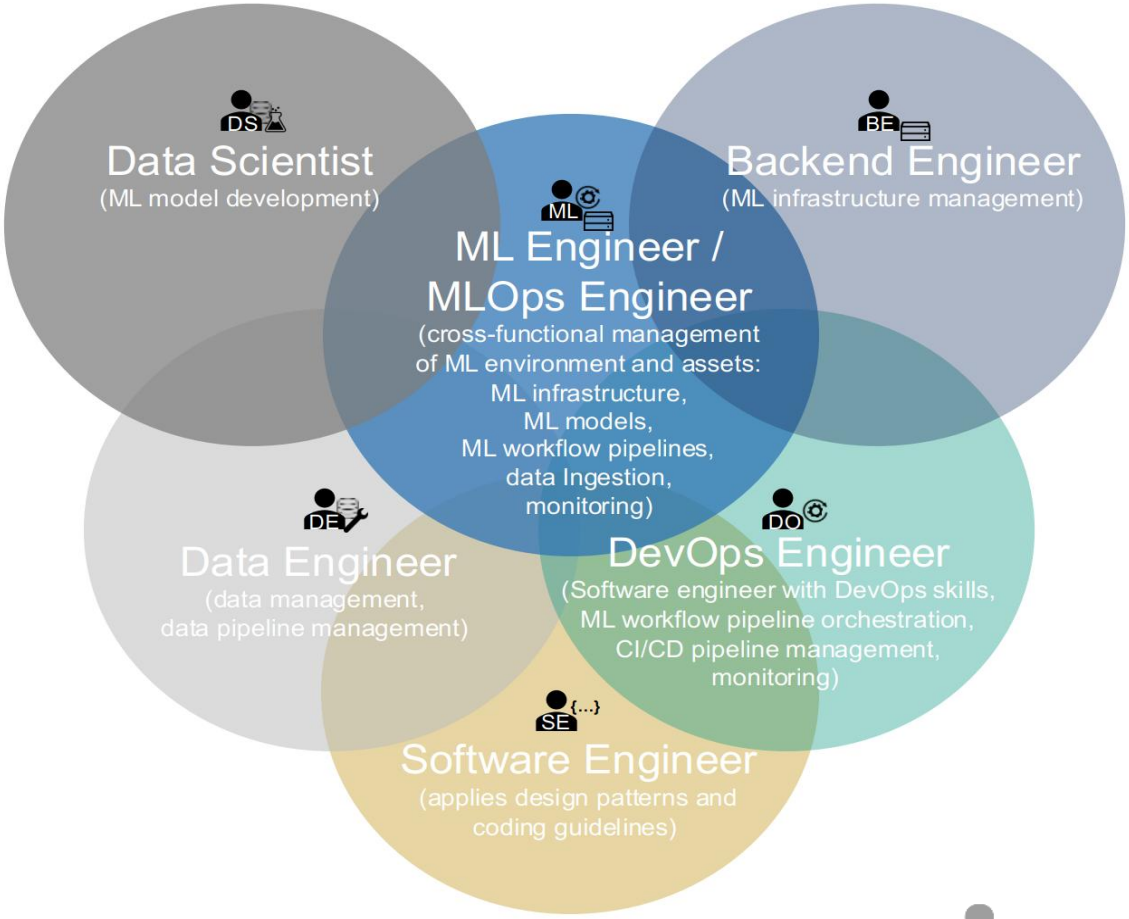
- Big data engineer & architects: create the underpinning software architecture; design, build, and manage the infrastructure and scalable data management systems that data scientists need to perform their analysis; outline business objectives and transform them into data-processing workflows; can be found across industries.

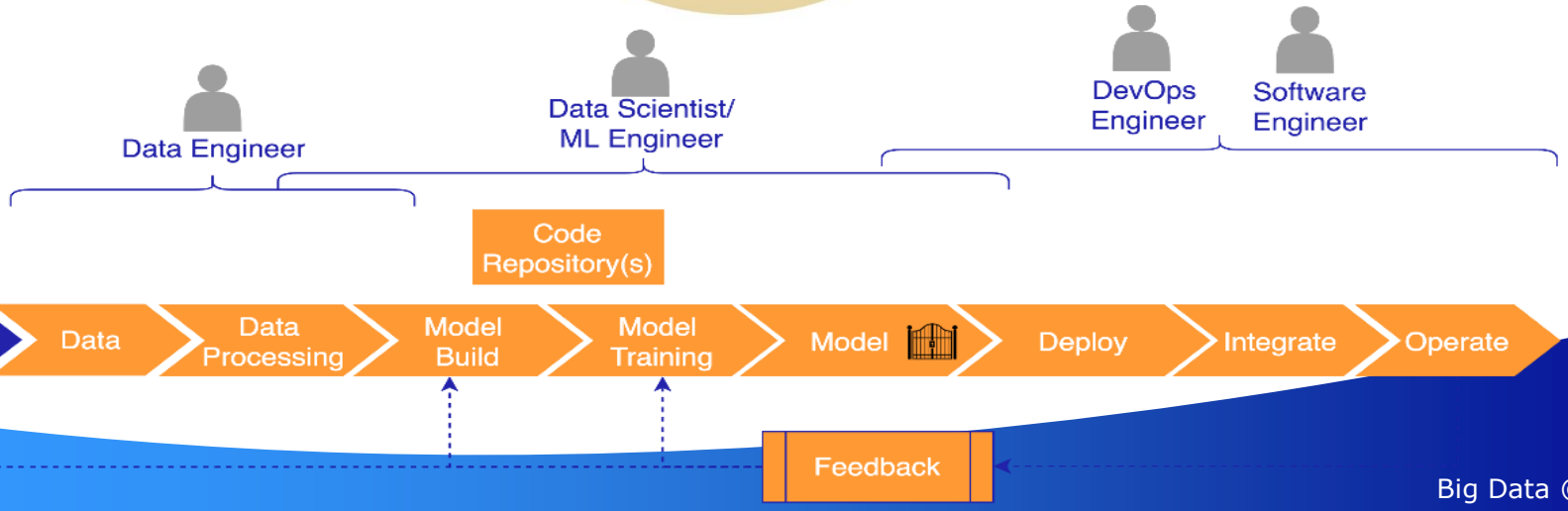
- Big data developers: apply their deep understanding of technologies such as Hadoop and Apache Spark with programming languages such as Java, Python and Scala to process data. By drawing on deep proficiencies in functional programming paradigms, they can effectively ingest data into broader big data platform ecosystems.

Big Data Jobs…

- Big data analysts: detect and analyze actionable data, such as hidden trends and patterns. By fusing these findings with their in-depth knowledge of the market in which their organizations operate, they can help leaders formulate informed strategic business decisions.

- Big data specialists: interrogate, ingest, analyze and transform complex sets of data. This ensures the necessary data is made available to the other team members who use it to uncover actionable insights and provide recommendations to improve business outcomes.
- …


DS
Data Scientist
(ML model development)
BE
Backend Engineer
(ML infrastructure management)
ML
ML Engineer /
MLOps Engineer
(cross-functional management
of ML environment and assets:
ML infrastructure,
ML models,
ML workflow pipelines,
data Ingestion,
monitoring)
DE
Data Engineer
(data management,
data pipeline management)
DO
DevOps Engineer
(Software engineer with DevOps skills,
ML workflow pipeline orchestration,
CI/CD pipeline management,
monitoring)
SE
Software Engineer
(applies design patterns and
coding guidelines)


Data Engineer
Data Scientist/
ML Engineer
DevOps
Engineer
Software
Engineer
Code
Repository(s)
Project
Identified
Data
Data
Processing
Model
Build
Model
Training
Model
Deploy
Integrate
Operate
Feedback

Skills required for Big Data Analytics

- Store and process
  - Large scale databases
  - Software Engineering
  - System/network Engineering
- Analyse and model
  - Reasoning
  - Knowledge Representation
  - Multimedia Retrieval
  - Modelling and Simulation
  - Machine Learning
  - Information Retrieval
- Understand and design
  - Decision theory
  - Visual analytics
  - Perception Cognition

